

Authentication System Architecture

MFA, Providers, & Keycloak Implementation Guide

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This document details the pricing, third-party providers, and specific Keycloak configuration steps required to set up robust authentication systems.

1. Executive Setup & Cost Matrix

Method	Recommended Providers	Estimated Cost	Keycloak Integration Type
Email OTP	Amazon SES, Resend, Brevo	~ ₹0.008	Native (SMTP Configuration)
TOTP (App)	Google Authenticator, MS Auth	Free	Native (OTP Policy)
SMS OTP	MSG91, Twilio, Fast2SMS	₹0.12 - ₹0.20	Custom SPI (Plugin Required)
Push (App)	Firebase Cloud Messaging	Free	Custom SPI (FCM Integration)

2. Third-Party Provider & Keycloak Integration Guides

A. Email OTP Integration

3rd Party Provider: Amazon SES (Cost-effective for high volume) or Resend (Developer friendly).

Keycloak Setup Steps:

1. Obtain SMTP credentials (Host, Port, Username, Password) from your provider.
2. In Keycloak, navigate to **Realm Settings** → **Email**.
3. Enter your provider's SMTP details (e.g., Host: `email-smtp.ap-south-1.amazonaws.com` , Port: `587`).
4. Enable **Authentication** and enter your credentials. Save and test the connection.
5. Navigate to **Authentication** → **Flows**, copy the standard Browser flow, and add the **Email OTP** execution block as a 'Required' or 'Alternative' step.

B. Authenticator App (TOTP) Integration

3rd Party Provider: None required for backend. Users download Google Authenticator, Microsoft Authenticator, or FreeOTP on their devices.

Keycloak Setup Steps:

1. Navigate to **Authentication** → **Policies** → **OTP Policy**.
2. Verify default settings: Type = **TOTP**, Algorithm = **SHA1**, Digits = **6**.
3. Go to **Authentication** → **Flows**. Duplicate the default Browser flow.
4. Find the **OTP Form** execution step and change it from 'Disabled' to **Required**.
5. Bind this new flow as the default Browser flow.
6. To force setup, go to **Authentication** → **Required Actions** and enable **Configure OTP**, setting it as default for new users.

C. SMS OTP Integration

3rd Party Provider: MSG91 (Best for Indian DLT compliance) or Twilio (Global reach).

Keycloak Setup Steps:

1. Download an open-source SMS Provider `.jar` file (e.g., `keycloak-sms-provider`) compatible with your Keycloak version.
2. Place the `.jar` file into the `/opt/keycloak/providers/` directory on your server.
3. Run the build command (`kc.sh build`) and restart the Keycloak service.
4. In the Keycloak Admin Console, a new **SMS Provider** settings tab will appear under Realm Settings.
5. Enter your MSG91/Twilio API Base URL, Auth Key, Sender ID, and your approved DLT Template ID.
6. Go to **Authentication** → **Flows** and add the newly available **SMS Authentication** step to your duplicated Browser flow.